

LifeLens

Multimodal Emergency Response Analytics Device

Isha Haider | Kameel Kasumu | Aryan Karadia | Saba Sadoughi | Caleb Oviawe | Aly Mohamed
Academic Advisor: Yani Ioannou

Motivation

In the most chaotic moments of a medical emergency, paramedics must save lives while **mentally** tracking every medication, every injury, and every intervention; information that will only be communicated later under equally **intense pressure**.

This is not a failure of the responders; it is a **limitation of the system**. No human can perform at peak clinical capacity while also acting as a real-time recorder. When critical details are lost, **the patient pays the price**.

Surgeons often receive only **fragments** of the full picture, with critical details uncovered only once the patient is on the table, **compromising preparation time**.

Solution

LifeLens is a **body-worn** device designed for paramedics during trauma response.

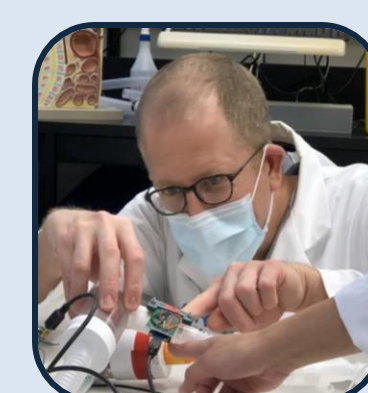
Through synchronized audio and video streams, it continuously **captures** and **extracts critical** information, including injuries, interventions, and medications, then converts it into structured, **time-stamped updates** that can be quickly understood at a glance.

These updates are delivered to hospitals and surgeons in near real-time, enabling earlier preparation for the patient's **most critical** injuries. The result is **faster** communication, **better-informed clinical decisions**, and stronger continuity of care from field response to surgery.

Who Do We Help?



Paramedics focus on saving lives and spend more time in the field.



Hospitals and surgeons receive accurate, time-stamped updates that help them prepare for the patient's most critical injuries before arrival.



Patients experience faster, smarter decisions that improve survival.

Jetson Orin Nano

Raspberry Pi Arducam IMX219 8MP, V2.1

ReSpeaker 4-Mic Array (Seeed)

Custom 3D Enclosure

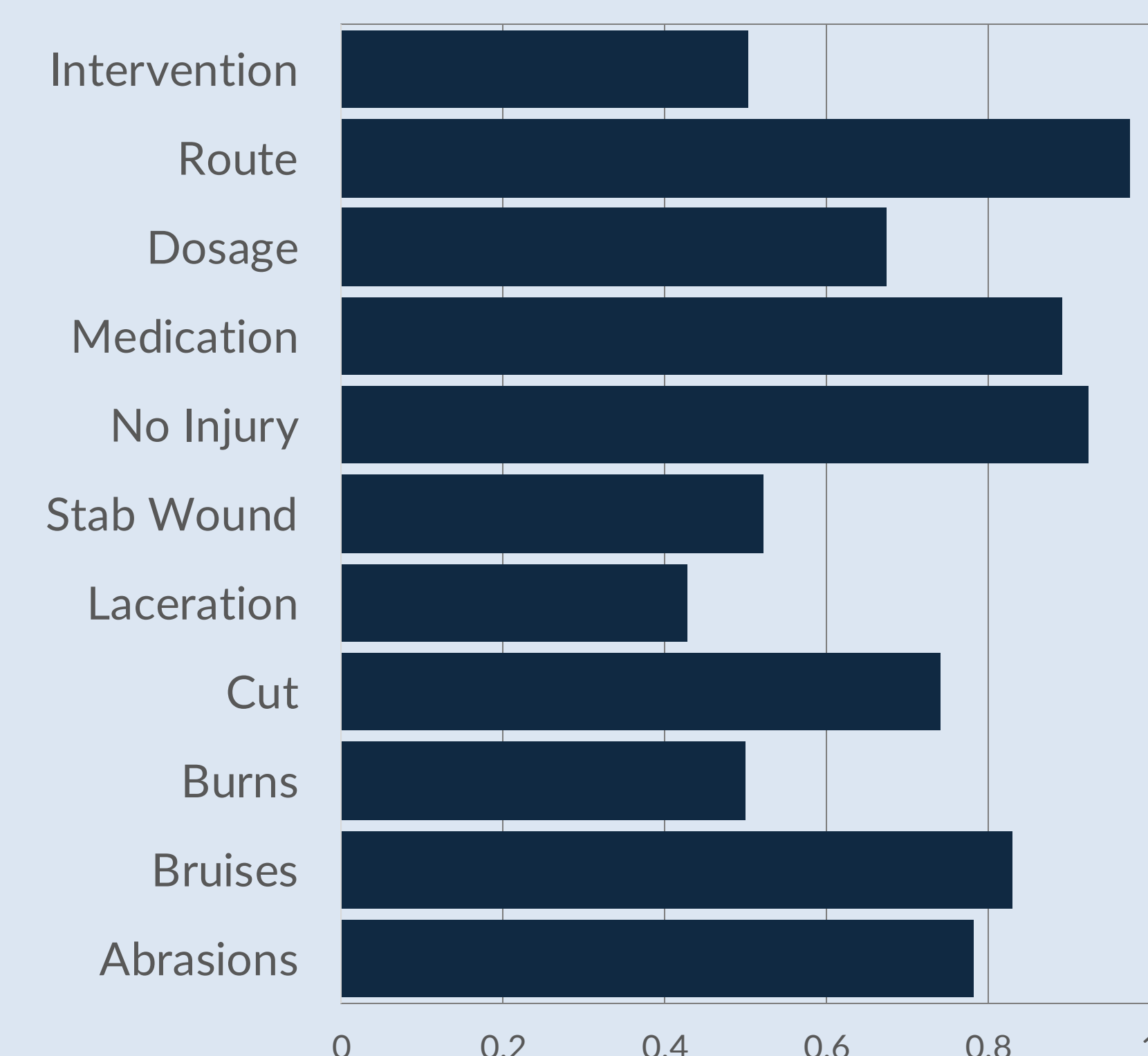
LED

Button

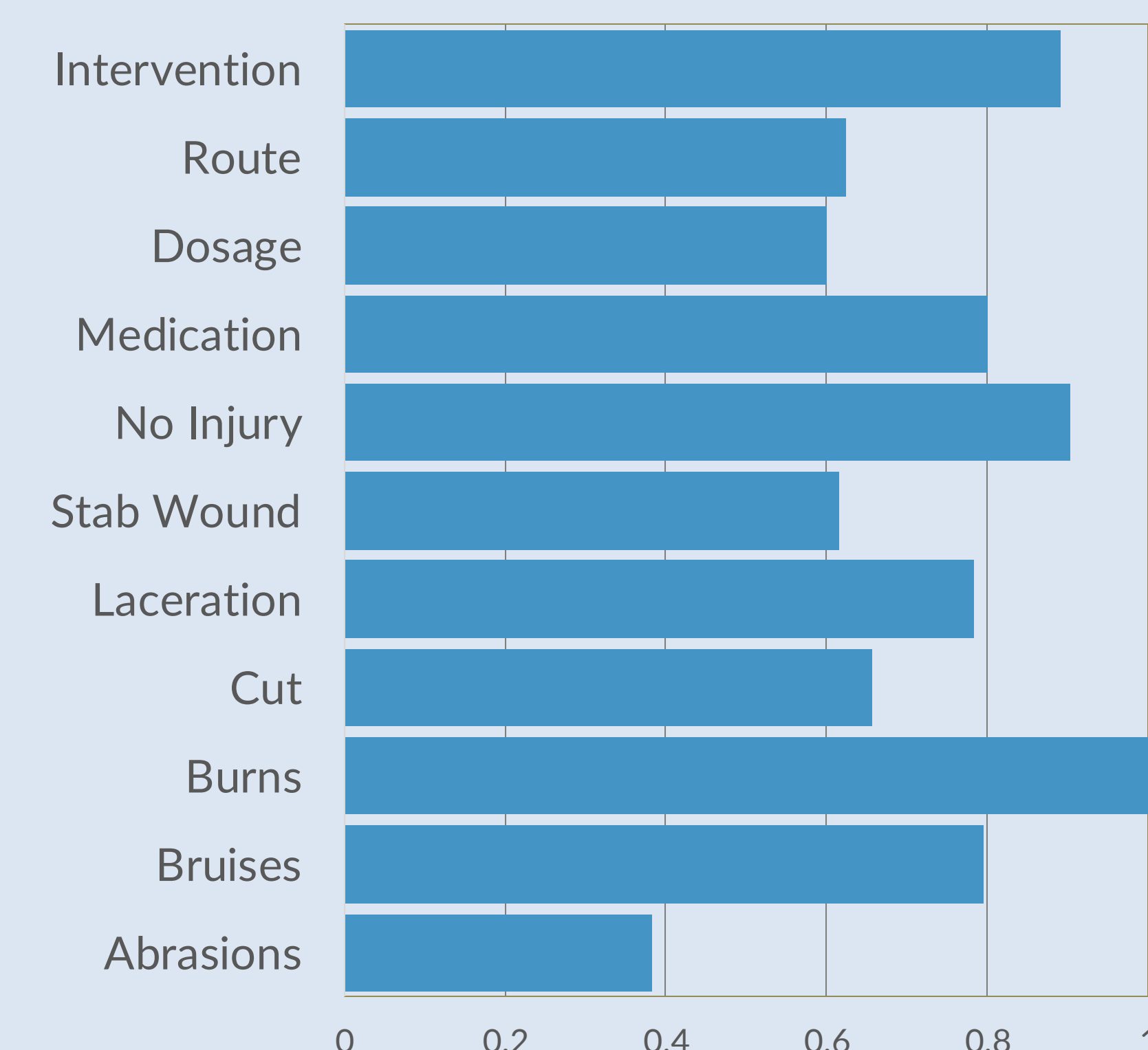
Current Gaps in Trauma Care

- 40% Information lost during paramedic handoff
- 60m Critical survival window for trauma patients
- 30s Time to relay information to trauma team
- 1/3 Paramedic spends time on documentation

Precision Metrics



Recall Metrics



To validate LifeLens, we conducted **multiple live trauma simulations** of emergency scenarios at **Foothills Hospital** with **8 trauma doctors**.

One dataset collected was used to **refine system performance**, while a separate validation set measured transcription accuracy, medication extraction, and intervention identification under **deployment-like conditions**.

DATA ACQUISITION



Microphone Array
Real-Time Audio Stream

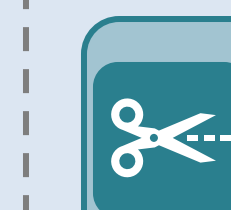


Stereo Camera
Continuous Video Stream

queued audio chunks + image frames

LIFELENS EDGE PROCESSING

AUDIO PIPELINE



Chunk Audio



Transcribe
Whisper



Anonymize
PII Redaction



Custom NLP
medCAT & med7

VIDEO PIPELINE

OBJECT TRACKING FOR PATIENT



Marker Detect
April-Tag Marker



Enroll Patient
Feature Embedding



Re-Identification
Nvidia ResNet50



Body Part Detection
Ultralytics YOLOv8



Injury Classification
Swin-Tiny Transformer

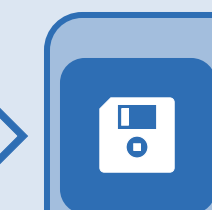
SECURE DATA TRANSFER



Data Transport
Tailscale VPN Tunnel

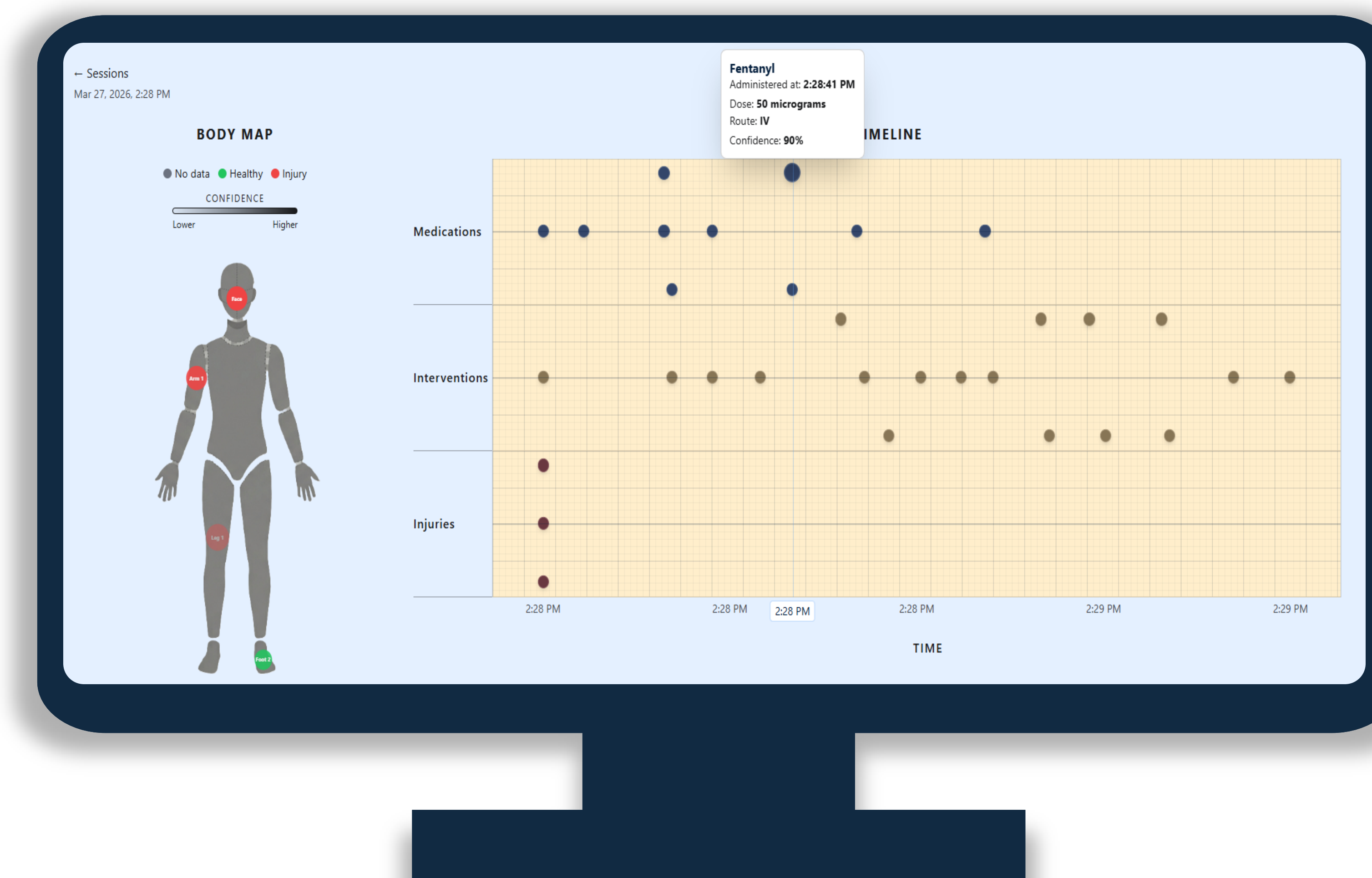


Encrypt Images
Symmetric Key



30-Day Storage
SQLite

USER INTERFACE



What's Next?

1

More Injuries

2

Partial Re-ID

3

Real Time

4

Hospital Integration

5

Wireless Setup

6

Radio Transmission



UNIVERSITY OF CALGARY
CUMMING SCHOOL OF MEDICINE



SCHULICH
School of Engineering